

KHOA DANG CAO

BACK-END DEVELOPER · RESEARCHER

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Hi, Khoa here! I'm a back-end developer and researcher, currently in my internship phase and looking forward to moving into a full-time role. I have a strong enthusiasm for R&D, sharing ideas and finding smart solutions. I enjoy teamwork and always look for ways to grow. My long-term goal is to pursue higher education abroad.

EDUCATION	Birmingham City University <i>Bachelor of Science with Honours in Computer Science</i> <i>International Joint Program with University of Information Technology, VNU-HCM</i>	Birmingham, UK 2023 - 2026 (expected)
EXPERIENCE	MindX Technology School HCMC, Vietnam Mentor / Teaching Assistant for K12 Robotics and Coding Programs - Led trial classes to inspire students' interest and uncover their strengths. - Connected with parents to suggest learning paths based on students' abilities. - Observed and assisted students during lessons. DigiSER Digital Services Company HCMC, Vietnam Full-stack Developer (React - Django) - Researched and analyzed various types of Vietnam civil status documents. - Developed a data entry website for digitizing civil status documents. - Implemented field validation and standardization for accurate data input.	2024.12 - present 2024.04 - 2024.07
HIGHLIGHTED PROJECTS	Performance Evaluation of Machine Learning Algorithms for an Intrusion Detection System (IDS) using the CICIDS2017 Dataset <i>Role: Project Lead, ML Developer, Main Author</i> Tech stack: Python, Jupyter Notebook, LaTeX. - Implemented Intrusion Detection System using Random Forest (RF), Retrained RF, KNN and XGBoost for detecting DDoS, DoS, Botnet, PortScan and Web Attacks. - Evaluated models based on F1-score, training/testing time and their effectiveness against different attack types. - Authored a detailed report documenting methodology, experimental results and performance analysis of IDS models. Read the report here. KhaBanh.life: Personalized AI Hairstyle Try-On Website <i>Role: Project Lead, Backend Developer</i> Tech stack: Spring Boot, FastAPI, PostgreSQL, FastHairGAN, Uvicorn, Docker - Developed the user service backend using Spring Boot, managing authentication, account operations, function requests, balance tracking, etc. - Resolved deployment issues and integrated FastHairGAN with FastAPI, allowing users to apply hairstyles and hair colors to their own photos. Automated Digitization System for Vietnam Birth Certificates <i>Role: Project Lead, Backend Developer</i> Tech stack: Flask, PostgreSQL, VietOCR, Uvicorn - Developed a model for extracting and digitizing information from Vietnamese birth certificates using VietOCR, enhancing accuracy through document field validation. - Built the backend with Flask, managing authentication, user accounts, document processing workflows; designed an API for uploading scanned birth certificates and retrieving structured data.	2024.12 - 2025.01 2024.12 - 2025.01 2024.07 - 2024.08

ACTIVITIES	Google Developer Group on Campus - UIT 2024.11 - present <i>Development team member, responsible for organizing and hosting tech-sharing sessions. Helped in some technology posts on the official fanpage of Google Developer Group on Campus - UIT, helping to share knowledge with the community.</i>
	May Tinh Cu - Tri Thuc Moi (Old Computers - New Insights) 2024.01 - present <i>A volunteer team since 2008 that repurposes and repairs used computer hardware to create new educational opportunities. Our slogan is "17 years of bringing knowledge to every corner."</i>
	Youth Union 2020.09 - present <i>Actively participated in Youth Union and social activities since high school. Served as the school's Youth Union Secretary for the 2021-2022 term. Currently involved with the Faculty of Computer Science Youth Union at UIT, though less actively.</i>
	Nguyen Huu Huan High School IT Club 2023.08 - 2024.06 <i>Former advisory member, organizing seminars, mentorship sessions and career orientation programs. Provided knowledge-sharing sessions to support students in their technical growth. Planned and authored educational content for 'Weekly Python,' a media program aimed at introducing high school students to Python programming.</i>
TECH-SHARING SESSIONS	UIT Google Developer Group on Campus - Git/GitHub Sharing 2024.11 <i>Prepared learning materials and delivered a sharing session for UIT Google Developer Group members on Git - GitHub fundamentals, workflow best practices and collaboration via branching and pull requests.</i>
	Trainee Program 2024 - Developer Starter Pack Sharing 2024.11 <i>Developed session materials and exercises. Presented the Developer Starter Pack session on Markdown and Git/GitHub for Trainee Program 2024, organized by the UIT Computer Science Faculty Youth Union, to equip university freshmen with essential skills.</i>
	Nguyen Huu Huan High School IT Club - Career Orientation Session 2023.11 <i>Participated in a career orientation session hosted by the NHH IT Club, sharing insights and experiences to help students navigate their university choices and career paths in IT field.</i>
AWARDS & HONORS	• Third Prize in Information Technology Track, ISE Spark of Ideas Fall 2024 2024.12 • Certificate of Merit, University of Information Technology Youth Union 2024.12 • Certificate of Recognition, Faculty of Computer Science Youth Union 2024.11 • Certificate of Appreciation, Binh Thuan Provincial Youth Union 2024.07 • Merit-Based Scholarship, University of Information Technology 2024.03 • Most Favourite Startup Project, UIT Creative Startup Ideas Hackathon 2023.12
SKILLS	Languages: Comfortable using English in both speaking and writing (IELTS 6.0, 2023). Tech Stack: Java (Spring Boot), Python (FastAPI, Flask, Django), C++, PostgreSQL, Docker. Tools: Git/GitHub, Postman, Adobe Creative Suite, Microsoft Office, JetBrains IDEs. Operating Systems: Windows, macOS, Ubuntu (Desktop and Server). Soft Skills: Public Speaking, Risk Management, Mentorship. Others: Graphic Design, Content Creation, Computer Repair & Troubleshooting.